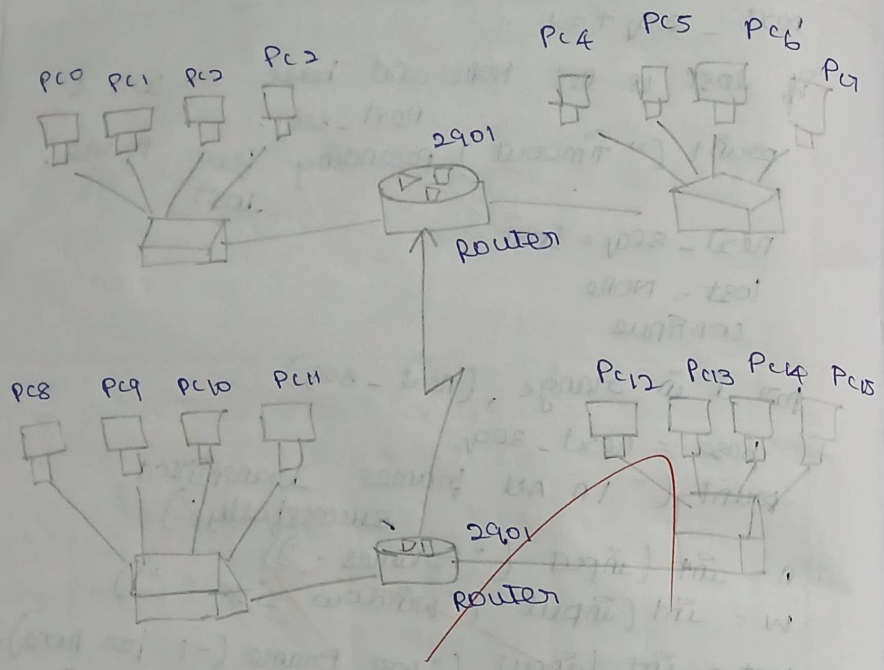


22/8/26

EXPERIMENT-9

IMPLEMENTATING OF SUBNETTING IN CISCO PACKET TRACER

Aim:

To implement subnetting in cisco packet tracer simulator.

Requirements:

* Cisco Packet Tracer
* Routers
* Switches
* PCs
* Ethernet cables

Procedure:

1. Create Network Topology
   * Open Cisco packet Tracer.
   * Add routers, switches and PCs.
   * Connect all devices using ethernet cables.
2. Perform Subnetting
   * Network : 192.168.1.0/24
   * Subnet mask : /27 (255.255.255.224)
   * Number of subnets : 8
   * Usable hosts per subnet : 30

IP Address configuration

Devices	IP Address
R1 G0/0	192.168.1.1
R1 G0/1	192.168.2.1
	192.168.1.11

Testing the network

open a command prompt on each Pc and try to ping the other Pc

Fire	Last status	Source	Dest	Type	Color	Time (sec)	Periodic	Num
○	Successfull	Pc4(2)	Router	ICMP	■	0.000	N	12
○	Successfull	Pc4(2)	Pc1(1)	ICMP	■	0.000	N	13
○	Successfull	Pc0	Router0	ICMP	■	0.000	N	14

Pc2	192.168.1.12
Pc3	192.168.1.13
Pc4	192.168.1.14
Pc5	192.168.1.15
R2 Fo/0	192.168.3.1
R2 Fo/1	192.168.4.1

Router configuration

enable
configure terminal
interface FastEthernet 0/0
ip address <IP address> <subnet mask>
no shutdown
exit.
ip address <IP address> <subnet mask>
no shutdown
exit.

switch configuration

enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit.

Pc configuration:

Assign the required IP address, subnet mask, default gateway and DNS server to each Pc.

Testing:

Use the ping - command from the Pc

command prompt to verify connectivity between PCs and routers

observation:

Subnetting was successfully implemented using a /24 subnet mask, and connectivity between the devices was verified using ping.

Result:

Thus, subnetting was successfully implemented in Cisco packet tracer and network connectivity was verified.